

Arm	Group	Head	Loss	Features	Attention	Split	Status	Forecast	Calib. dev	Calib. cohorts
q_age_set	covariate	quantile	QuantileLoss	age at draw, care setting	self masked, cross	sequence	trained	yes	yes	yes
334f7e21	covariate	quantile	QuantileLoss	none beyond sex and analyte	self masked, cross	sequence	trained	yes	–	yes
q_age	covariate	quantile	QuantileLoss	age at draw	self masked, cross	sequence	trained	yes	yes	yes
q_set	covariate	quantile	QuantileLoss	care setting	self masked, cross	sequence	trained	yes	yes	yes
q_co	covariate	quantile	QuantileLoss	same-draw analytes, history	self masked, cross	sequence	trained	yes	yes	yes
q_age_co	covariate	quantile	QuantileLoss	age at draw, same-draw analytes, history	self masked, cross	sequence	trained	yes	yes	–
q_set_co	covariate	quantile	QuantileLoss	care setting, same-draw analytes, history	self masked, cross	sequence	trained	yes	yes	–
q_age_set_co	covariate	quantile	QuantileLoss	age at draw, care setting, same-draw analytes, history	self masked, cross	sequence	trained	yes	yes	yes
q_co_q	covariate	quantile	QuantileLoss	same-draw analytes, history and query	self masked, cross	sequence	trained	yes	yes	–
p_base	patient split	quantile	QuantileLoss	none beyond sex and analyte	self masked, cross	patient	trained	yes	yes	–
p_co	patient split	quantile	QuantileLoss	same-draw analytes, history	self masked, cross	patient	trained	yes	yes	–
p_causal	patient split	quantile	QuantileLoss	none beyond sex and analyte	causal: self and masked	patient	trained	yes	yes	–
p_full	patient split	quantile	QuantileLoss	all analytes, every past draw	causal: self and cross masked; ≤ 128 draw tokens	patient	trained	yes	yes	–
pa_k5	prior-anchored	quantile	QuantilePriorLoss (anchor, k=5)	age at draw, care setting	self masked, cross	sequence	started, no checkpoint	–	–	–
pa_k20	prior-anchored	quantile	QuantilePriorLoss (anchor, k=20)	age at draw, care setting	self masked, cross	sequence	partial, epoch 0	–	–	–
pa_tau	prior-anchored	quantile	QuantilePriorLoss (anchor, k=5, tau=365d)	age at draw, care setting	self masked, cross	sequence	partial, epoch 0	–	–	–
pf_k5	prior-anchored	quantile	QuantilePriorLoss (floor, k=5)	age at draw, care setting	self masked, cross	sequence	partial, epoch 0	–	–	–
pg_k5	prior-anchored	gate	QuantilePriorLoss (gate, k=5)	age at draw, care setting	self masked, cross	sequence	started, no checkpoint	–	–	–
pn_k5	prior-anchored	nig	StudentTNLLLoss (nu0=5)	age at draw, care setting	self masked, cross	sequence	partial, epoch 0	–	–	–
gk_k5	prior-anchored	gaussian	NORMALoss (k=5, lambda=0.1, weighted by n)	age at draw, care setting	self masked, cross	sequence	partial, epoch 0	–	–	–

Table 1: Every NORMA arm trained since the covariate ablation began. All are NORMA2 at $d_{\text{model}} = 64$, 4 heads, 8 layers and 3 states, trained on EHRSHOT and MIMIC-IV together. *Attention*: the decoder layers are called with memory equal to target, so the causal mask applies to self-attention only, leaving the cross-attention block bidirectional over history, unless `--causal_memory` masks both.